

DEVELOPMENT OF AUTOMATED COMBINED MULCHING SHEET LAYING AND FERTILIZER APPLICATION ROBOT

A Project Report

Submitted by

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in

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under the supervision of

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Certificate

This is to certify that this project report titled **“Development of Automated Combined Mulching Sheet Laying and Fertilizer Application Robot”**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, by **Anjjai Rubhak T J, Adaikkappan S, Vishal Srivatsan S** is a bonafide record of work carried out by them, under my supervision and that it has not been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.

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We do hereby declare that this project report titled “**Development of Automated Combined Mulching Sheet Laying and Fertilizer Application Robot**”, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, is a true record of work carried out by us, under the supervision of **Mr. Peeyush K P** and that all information contained herein, which do not arise directly from our work, have been properly acknowledged and cited, using acceptable international standards. Further, we declare that the contents of this report have not been submitted, in part or in full, for the award of any other degree or diploma.

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Abstract

Modern farming practices have led farmers to carry out practices such as fertilizing and mulching manually. This process is labor-intensive, time-consuming and often requires different machines, making it costly and complicated. Although robotic technology exists in such farming practices, existing ones are mostly not multifunctional and are expensive which are suitable for the use of industrial-scale farming. In order to improve on these, this project is proposing to create an automated robot that can carry out two processes simultaneously: fertilizing and mulching sheet laying.

The proposed system integrates mechanical design, electronics, and software components for the synchronized fertilizer dispensing and mulching. A differential drive mechanism powered by DC motors and controlled through Raspberry Pi 4 ensures smooth navigation across crop beds. The mechanism for the application of fertilizers is sliding gate for accurate application. Similarly, the mulching mechanism is rollers along with a pressing system for efficient laying of the sheets without any displacement. As for feedback, a Hall-effect sensor is put on the wheels to find the speed and displacement to control the robot. Moreover, a web application is also proposed for efficient monitoring and controlling of the robot for the application of fertilization and mulching.

The expected outcome of the project will be a cost-effective, user-friendly robotic system that can reduce the labor and input costs of the small-scale farmers while increasing the efficiency of the operation. The fertilizer and mulching system will not only reduce the operation cost, but it will also help in the better management of the soil as well as the yield. The project presents a novel application in the field of agricultural robotics, navigation, and web-based control.

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List of Abbreviations

4WD	Four-Wheel Drive
BCM	Broadcom
BWD	Backward
CAD	Computer-Aided Design
FWD	Forward
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
GUI	Graphical User Interface
HMI	Human-Machine Interface
LED	Light Emitting Diode
PWM	Pulse Width Modulation
BMS	Battery Management System

List of Symbols

V	Voltage
A	Ampere
Ah	Ampere-Hour
W	Watt
Wh	Watt-Hour
$N.m$	Newton-metre
$^{\circ}$	Degree
t	Time
dt	Time interval
m	Meter
kg	Kilogram
g	Gram
g/s	Gram per second
m/s	Meters per second
Hz	Hertz
$\%$	Percentage
L	Row Length in meters
S	Row spacing in meters
R	Number of rows
ϑ	Servo motor angle
v	Velocity in m/s

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1. Introduction

One of the oldest, yet most important, human activities has always been agriculture, which has always been the backbone of civilization, yet continues to play an important role in the food security and economic development of countries, especially developing ones. Nevertheless, despite all the advancements that modern technology has brought, farming is still being done manually, making it less efficient, costly, and less productive.

There are many numerous activities involved in agriculture, the application of fertilizers and mulching stand out as two very important ones. Fertilization plays the crucial role of supplying the crops with the nutrients they require to grow strong and healthy. Mulching ensures the retention of water in the soil, controls the temperature, prevents erosion, and prevents the growth of weed also. However, the two activities have proved to be very inefficient and labor-intensive. Fertilizers applied to the crops through manual application tend to be very random, resulting in wastage and pollution. Mulching, on the other hand, requires a lot of time and effort. But there is a great possibility of the sheet being misplaced or even torn. Since there are two distinct machines for these activities, the farmer would need to invest in both, leading to higher costs and increased time consumption.

This is really tough for small and marginal farmers, who are a significant portion of the country but don't have the means to buy large, costly systems. What they really need is a machine that is small, versatile, affordable and also that can reduce their dependence on labor while still improving efficiency. Unfortunately, majority of the available agricultural robots currently in the market are intended for use in large farms. They are expensive, hard to use, making them unsuitable for smallholders who require more integrated and user-friendly options.

To address this gap, the Idea of a Fertilizer Application and Automated Combined Mulching Sheet Laying Robot is proposed. The goal of this system is simple but impactful: to bring together two essential farming operations into one platform, saving farmers time, effort, and money, while making the process more accurate and sustainable.

1.1 Motivation

The motivation for this project comes from the need to support small-scale farmers with affordable solutions, as they continue to face rising labor costs, a shortage of workers, and the expense of purchasing materials.

1.1.1 Challenges in Current Farming Practices

One of the major issues in the current system of farming is the lack of consistency in the manual process. The application of fertilizers is not consistent when done manually. Some areas end up receiving excess fertilizers, while others receive less than required. This not only wastes the fertilizers but also leads to salinization of the land, groundwater pollution, and a decrease in crop yields. Similarly, the process of mulching is also not efficient as the sheets have to be manually carried, unrolled, and pressed onto the land. If not done properly, it leads to air pockets in the sheets, resulting in water loss and the growth of weeds. Moreover, if the sheets are not properly aligned, they get dislodged either due to wind or water. Mulching is done after fertilization, so both processes have to be done efficiently.

1.1.2 Limitations of Existing Agricultural Robots

While mechanical solutions do exist, such as fertilizer spreaders and mulching machines, but they are used in large-scale farming. The equipment is big, expensive and are heavy to be used in small-scale farms. Moreover, the equipment is mostly single-function devices, which means that a farmer would need to purchase separate equipment to handle fertilizer application and mulching, respectively. However, the improvement in technology has brought us some benefit in terms of solutions but are also not budget-friendly for these farmers. For example, agricultural robots that use GPS, sensors can navigate the fields. However, the accuracy of these robots makes them expensive and also the farmer has to trained in this.

1.1.3 Need for an Integrated Solution

Given these reasons, it is evident that there is an urgent need to create an integrated system that is able to accomplish the tasks of fertilizer application and mulching in one go. This is because such a robot would not only save time and labor but also help in the uniform application of fertilizer, reduction of fertilizer wastage, and the creation of mulch sheets that help in the conservation of water, reduction of weeds, and the regulation of temperature in the soil. This is exactly what this project is trying to achieve: the creation of an affordable and two-in-one robot that is able to accomplish the tasks of fertilizer application and mulching in one go.

1.2 Objective

The aim of this project is to design and develop a robotic system that counters the challenges of the conventional farming methods through the integration of the application of fertilizers and the process of mulching into a single system. The initial step of this project is to design a chassis that is compact, strong, and capable of holding all the electronic parts, motors, and other mechanical parts of the robot while ensuring that the structure is light, strong, and adaptable to the varied conditions of the soil and the fields where the robot would be used. An integral component of the robot would be the mechanism of fertilizer application, which would be a conical hopper made of sheet wood and a sliding gate that would precisely allow the application of small amounts of fertilizer through the movement of the robot on the fields. At the same time, the mechanism of mulching would be developed through the integration of rollers, spindles, supports, and pressing mechanisms to ensure that the sheet is tightly pressed to prevent the formation of air pockets and the movement of the sheet through the wind or water.

Another crucial goal is the integration of electronics and power systems, where a Raspberry Pi centrally controls batteries, motor drivers, DC motors, and servo motors to ensure smooth operation. A web-based application that allows farmers to remotely adjust field parameters like length, width, and fertilizer dosage while simultaneously tracking the robot's status in real time will be developed to further improve usability.

1.3 Novelty

The novelty of this project lies in its unique combination of dual functionality, affordability, and ease of use, which have been specially designed to the requirements of small/marginal farmers. While other machines have been designed to perform a single operation, the proposed robot has been designed to perform both fertilizer as well as mulching sheet laying operations simultaneously, thus reducing the time, effort, and overall cost of operation. Additionally, the proposed robot would be equipped with a special web application, which would allow the farmers to operate the robot, start the operation, set the dimensions of the field, as well as control the amount of fertilizer to be used, without the need to be technically skilled. Most importantly, the proposed project has been designed to be simple, cost-effective, as well as practical, which would be beneficial to the small/marginal farmers, as they would be able to afford the machine, which would otherwise be out of reach due to the high cost of the commercially available robots, thus providing a practical solution to the challenges associated with the operation of such machines in small-scale farming.

2. Literature Survey

The use of robotics in agriculture is emerging as one of the most promising applications of technology in farming, with uses ranging from planting and spraying to harvesting, mulching and fertilization. While research often highlights the efficiency and optimization these systems can bring, most of them are still too expensive, hard to use, or limited to a single function only. This makes them less practical for small and marginal farmers who could benefit the most from such innovations [1].

When it comes to mulching systems, most of the research so far has focused mainly on mechanical designs that ensure proper sheet placement. For example, Yughander et al. [2] proposed an automatic mulching machine that uses rollers and pressing mechanisms to align the sheet tightly with the soil so that there is no displacing of the sheet. Similarly, Li et al. [3] developed a potato seeding machine and also integrated it with mulching. These studies show us the importance of applying tension to the sheet to ensure good soil contact, but they do not combine this process with fertilization, which is addressed shortly.

Research on multifunctional agricultural robots also shows clear advantages. Munna et al. [4] developed a robot that could seed and apply nitrogen fertilizer at the same time, which led to better efficiency and yield. Although this work did not include mulching and it very expensive, it shows the benefits of combining different kinds of processes into one platform.

Navigation is another important factor. High-end GPS and vision systems are accurate, but their cost makes them unaffordable for small-scale farmers. More affordable ones like odometry, which is based on encoders and dead reckoning, have been shown to work well in row-based field operations. Ferreira et al. [5] and others [6] argue that wheel encoders are suitable for controlled movements, even though they suffer from issues like drift and wheel slip.

Ease of use is also important. Control systems which are complex to use often discourage farmers from using these technologies, whereas simple web-based interfaces are much more farmer-friendly. Recent studies [1], [2] highlight that giving farmers the ability to easily set parameters and monitor the robot through a web app is a promising c. In line with this, this project includes a Flask and JavaScript based interface for monitoring and remote control.

From the review of existing works, it is clear that although systems exist for both mulching and fertilization, they are usually distinct and rarely designed for small-scale farmers. Not many try to combine both functions into one platform. The proposed robot addresses this gap. This makes it a practical and affordable solution that could benefit small and marginal farmers.

3. Methodology

The methodology outlines the procedure followed to design, develop, and validate the proposed robotic platform. It involves multiple stages from problem analysis and designing to fabrication, software logic, integration and on-field testing.

3.1 Problem Definition

Traditional farming practices such as fertilizer application and mulching are still performed manually or with separate machines. Manual fertilizer application often leads to uneven distribution, wastage of inputs. Similarly, manual mulching is labor-intensive, time-consuming and prone to errors such as loose placement and tearing of sheets. While robotic solutions are available, they are expensive and perform only one task at a time. This creates a gap for small and marginal farmers, who require a low-cost, compact multifunctional system which does both fertilizer application and mulching in a single operation. The aim is to design the platform that integrates both and reduces labor dependency, ensuring efficiency and remains affordable and user-friendly. This in turn reduces the soil exposure by reducing the number of times the soil is exposed to human intervention which helps in reducing the loss soil nutrients.

3.2 Flowgraph

The flowgraph of this project is shown in the Fig 3.2, all of the topics in the flowgraph are detailed in the upcoming pages.

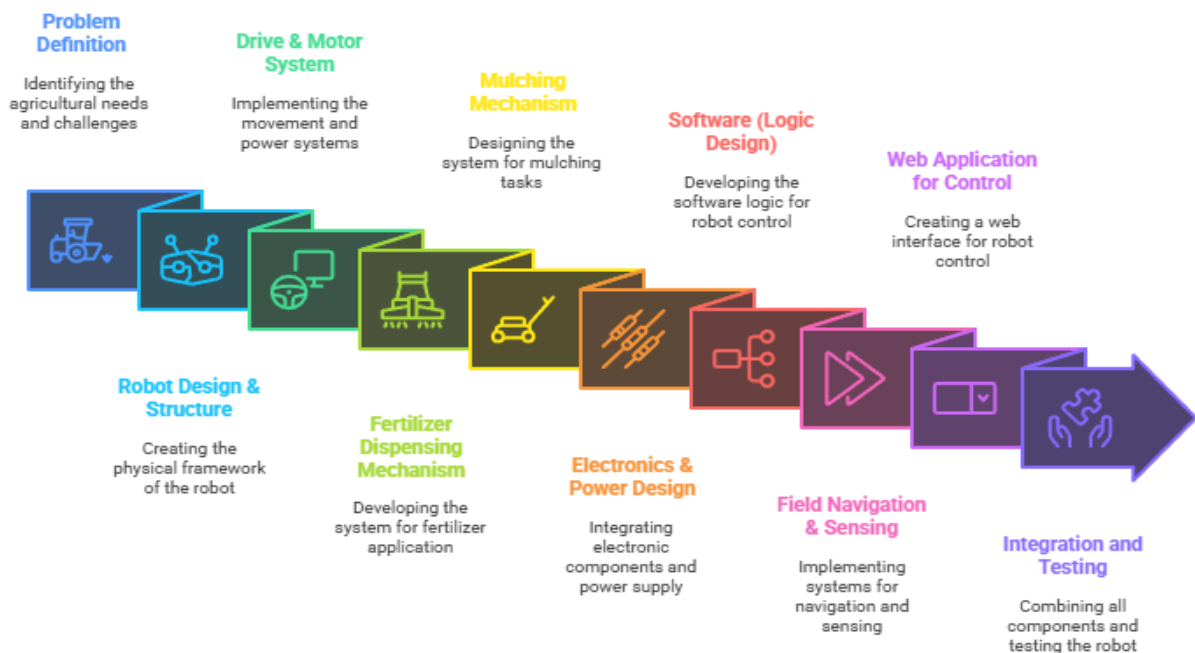


Fig. 3.2 The Flowgraph

4. Mechanical Design and Implementation

The mechanical design is a crucial component of the project, as it integrates the fertilizer dispensing unit, mulching mechanism, and chassis into a one platform. The methodology followed for the design is followed below.

4.1 Chassis Design

This section forms the structural foundation of the platform which is to be built. A light-weight frame was designed using Fusion 360 CAD to ensure the right fabrication of the intended design. This platform is designed to handle the placing of the motors, electronics, fertilizer hopper, and mulching roll holder (refer Fig 4.1). The materials selected are based upon durability, weight considerations, and affordability which resulted in choosing mild steel and aluminum.

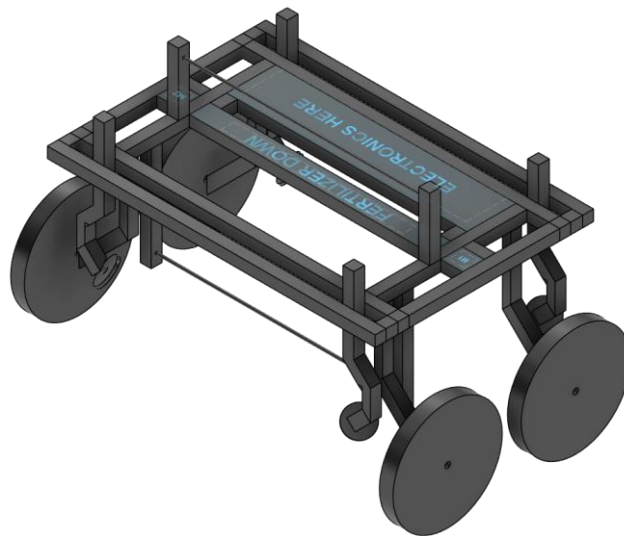


Fig. 4.1 The Designed Model

4.2 Hardware Implementation

The hardware implementation of the system that was proposed includes the mechanical design and the electronic parts that are required for the fertilizer dispensing, mulching operation, and robot movement. The system is designed to provide stable mechanical operation, safe power distribution, and dependable motor control in tough field conditions also. The hardware is made up of the chassis, fertilizer dispensing unit, mulching

mechanism, power system, motor drive system, and control electronics. The mechanisms are described in detail in the following subsections.

4.2.1 Fertilizer Dispensing Mechanism

The fertilizer dispensing unit in the model is designed as an agricultural hopper, which works in a basic fashion. The hopper is designed at the top of the platform with sloped inner walls to allow granules of fertilizer to flow naturally towards the outlet. We use a sliding gate at the bottom to control the rate at which the fertilizer is dispensed. The gate is connected to a servo motor, allowing the operator to open it partially or fully depending upon the required fertilizer dosage.

As the robot moves forward, fertilizer drops directly from the hopper outlet into the soil. By controlling the sliding gate with robot's speed, uniform distribution can be achieved. This is mechanically simple, cost-effective making it highly suitable for small-scale farming.



Fig. 4.2 Fertilizer Dispensing Unit

4.2.2 Mulching Mechanism

The mulching mechanism is designed to unroll, guide, and press plastic mulch film onto the soil surface in a uniform manner. At the rear of the robot, a primary spindle-type holder is provided to mount the mulch roll. To keep everything aligned and also avoid wrinkles or uneven placement, support rollers are set along the sheet's path. These rollers help keep the sheet centered and manage the tension properly while it's laid down.

To make sure that mulch sticks to the soil well, a pressing mechanism is used. This usually includes pressing plates that are mounted behind the guide supports. As the film rolls under these rollers, a downward pressure is applied, which increases the contact with the soil. Initially, the soil needs to be pushed together to form a raised bed before adding fertilizer. After the mulching is done, we need human-intervention to put sand at the sides of the film so that it stays in place. After operation in a row, we need cut the film so that there is no tearing or wrinkling for the next row.



Fig. 4.3 Mulching Mechanism Unit

5. Electronics and Power Source

The electronics and power source system is the backbone of the platform, facilitating the coordination of the fertilizer dispensing unit, mulching system and drive system. The heart of the design is the Raspberry Pi microcontroller, which acts as the central processing unit to manage the motor control, navigation and communication to the web-based interface. The circuit designed is given and explained further.

5.1 Circuit Design

The circuit design outlines the complete power and control system for the automated combined mulching sheet laying and fertilizer dispensing system. The lithium-ion battery pack has a built-in integrated BMS which is the source that powers the motor controller via an emergency stop button. The motor controller drives the two brushed DC geared motors using PWM and direction signals received from the Raspberry Pi, which will control the manner in which the robot moves during the creation of the plant beds, fertilizer dispensing, and laying of the mulching sheet. The hall effect sensor stuck on the wheels will be used to sense the rotation of the wheels by producing depending on the manner in which the wheels rotate, which will be used to monitor the speed of the wheels using the Raspberry Pi, to control the manner in which the mechanisms move in a synchronized manner. The servo motor will be used to control the sliding gate using the signals sent by the Raspberry Pi, used to control the manner in which the fertilizer is spread. A power bank will be used to power the Raspberry Pi.

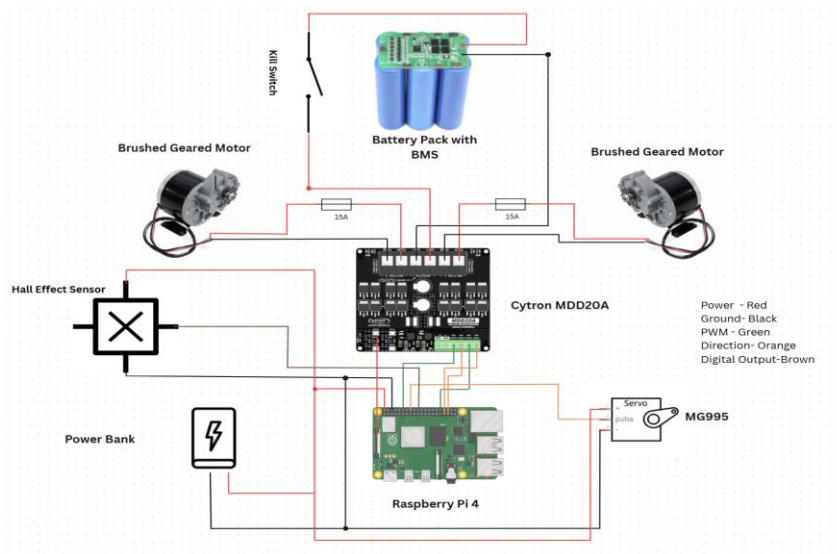


Fig. 5.1 Circuit Schematic

Table 5-1 Raspberry Pi pinout connections

S. No	Physical Pin	GPIO (BCM)	Connected To	Direction
1	Pin 1	-	Hall Effect Sensor Vcc	POWER OUT
2	Pin 6	-	Motor Controller GND	POWER OUT
3	Pin 9	-	Motor Controller GND	POWER OUT
4	Pin 11	GPIO 17	Hall Effect Sensor	IN
5	Pin 12	GPIO 18	Right Motor PWM	OUT
6	Pin 14	-	Fertilizer Servo Motor GND	POWER OUT
7	Pin 16	GPIO 23	Fertilizer Servo Motor PWM	OUT
8	Pin 18	GPIO 24	Fertilizer LED	OUT
9	Pin 20	-	Hall Effect Sensor GND	POWER OUT
10	Pin 32	GPIO 12	Left Motor PWM	OUT
11	Pin 33	GPIO 13	Left Motor Direction	OUT
12	Pin 35	GPIO 19	Right Motor Direction	OUT

5.2 Power Source

The lithium-ion battery pack with an integrated battery management system (BMS) is used. The battery outputs a voltage of 27.3 V with a capacity of 9Ah A, which is enough to power both the motor and electronics present. In this project, the battery supplies power to the motor controller for driving the DC motors. The BMS monitors the battery and protects it from undercharging, overdischarging and short-circuits, ensuring safe operation.

5.3 Motor Driver and Control Circuits

Brushed DC geared motors are used for robot movements because of their high torque capability, which is required for agricultural field conditions. Each motor operates at 24V DC with a power output of 250 W. These motors are responsible for driving the wheels of the robot, enabling forward and reverse movement during fertilizer dispensing, and mulching sheet laying operations. A dual-channel motor controller is used to drive these motors by receiving PWM and direction signals from the Raspberry Pi, therefore controlling the speed and distance based on the user input. The controlled motion ensures uniform fertilizer applications and proper laying of the mulch sheet.

Motor Specifications

- **Voltage: 24 Volt DC.**
- **Output: 250 Watt.**
- **RPM (after Reduction) – 300.**
- **Full load current – 13.4A.**
- **No load Current – 2.2A.**
- **Torque Constant – 8 N.m (80 kg-cm).**
- **Torque stall – 40 N.m (400 kg-cm).**
- **Sprocket: 9Tooth only fits #410 bicycle chains.**
- **For Chain Size: Pitch 0.5 inch.**
- **Roller Diameter 0.3 inch.**
- **Roller Width 0.16 inch.**
- **Weight : 2.46 kg**

Fig. 5.3a Specifications of the Motor

Item Type:	Motor Driver
Voltage Rating (V):	6 to 30
Current Rating (A):	20
No. of Channels:	2
Operating Voltage (VDC):	6 ~ 30
Input Logic Voltage(V)	1.8 V to 12 V
Peak Current (A)	60(10 sec)
Continuous Current (A):	20
PWM frequency	20 KHz
Length (mm):	89
Width (mm):	79
Height (mm):	2

Fig. 5.3b Specifications of the Motor Controller

5.4 Raspberry Pi controller

The core of the system is the Raspberry Pi , which basically acts as the brain of the robot, thus all major electronic operations are coordinated through it. It produces the PWM signals that control the speed of the drive motor and at the same time it manages the direction control logic for the two motor drivers, thus the steering can be done even at the minimum speed and with great accuracy. The Raspberry Pi also takes the input from the wheel encoders, thus allowing dead reckoning navigation which is a method for measuring the robot's position without the need for GPS. Besides this, it also controls the release function of the fertilizer dispensing unit that is a servo or a DC motor-driven device for the accurate release of the fertilizer. The Raspberry Pi, apart from hardware control, runs a Flask-based web application which is a Wi-Fi communication system that provides an easy interface for the farmer to monitor the robot, change the parameters and give the control commands at the time of execution.

5.5 Servo Motor and Hall Effect sensor

An MG995 servo motor controls the sliding gate of the fertilizer dispensing unit. It receives control signals from the Raspberry Pi and then slides the gate to the required amount of fertilizer during robot movements. This allows the robot to distribute fertilizer visually controlled and evenly while it is performing plant bed formation and mulching operations.

A hall effect sensor is used to monitor the movement of the robot wheels and provide feedback during operation. It produces pulses for each rotation of the wheel, which is then processed by the Raspberry Pi to measure the speed and displacement of the robot. This feedback is essential for maintaining a controlled speed during the fertilization process as well as the laying of the mulching sheets.

6. Software Logic and Control

The software and control system is the brain of the agricultural robot, Agribot. In this part, the software architecture of AgriBot system which includes the design of the concurrent control engine, guided navigation logic, fertilizer dispensing computation, and the web-based human-machine interface is studied. The Raspberry Pi Operating System environments facilitate the operation of all software programs on the Raspberry Pi 4. Moreover, the software is executed completely using Python 3. GPIO access is enabled by RPi. GPIO libraries. The web interface is served using Flask.

6.1 Software Architecture Overview

The firmware of the whole system is divided into 3 concurrent execution layers that operate during the run-time. A fixed 20 Hz rate thread which runs on dedicated hardware is responsible for all time-critical motor and servo control. When the operator starts a guided operation, a mission thread is triggered to handle the sequencing of row traversals, turns, and operator pause points at a higher level. Flask HTTP server deals with all the communication through the browser-based interface and runs on the main thread.

The three layers share a mutable state structure, which is a Python dictionary called state that includes all telemetry values, status flags, and mission parameters which is single threading. Lock() object protects access to this shared state throughout the application. This design was chosen, over many fine-grained locks because a single lock eliminates the class of deadlock errors that occur when one thread holds one lock and waits for another while the second thread does the opposite. The drawback is a slight reduction in locking accuracy, but at the relevant control frequencies, this has no impact.

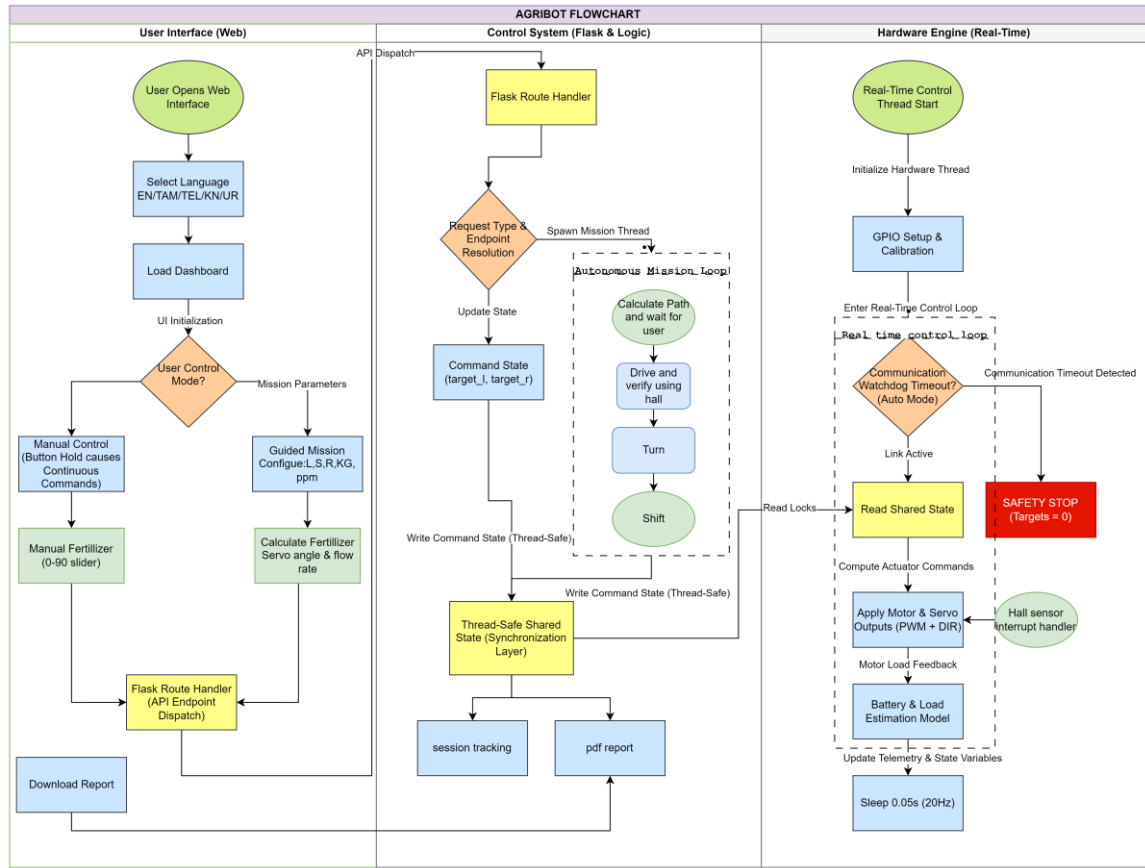


Fig. 6.1 Software Architecture Flowchart

The overall software flow is illustrated in Figure. 6.1. On startup, the hardware engine thread is launched as a daemon thread before the Flask server begins accepting connections. When an operator starts a guided mission through the web interface, a new mission thread is spawned and runs until the mission completes, aborts, or encounters a safety stop condition.

6.2 The Hardware Engine – Real-Time Control Loop

The hardware engine is the lowest layer of the software stack and the only component that directly interfaces with GPIO hardware. It executes as a daemon thread launched at system startup and runs an infinite control loop with a fixed period of $dt = 0.05$ s, corresponding to a control update rate of 20 Hz. Each iteration reads the current motor targets and applies the PWM ramp calculation. The output is then given to the GPIO motor drivers. The servo position and battery estimate are updated at the end of the same tick. GPIO is configured in BCM mode. The left motor uses pins 12 and 13 for PWM and direction respectively, the right motor uses pins 18 and 19. Pin 23 contains the 50 Hz servo signal and pin 24 drives the status LED. The Hall effect sensor is connected to Pin 17.

6.2.1 Motor PWM Ramping

During testing sudden PWM changes were observed and it was a problem. This is because the drivetrain would jerk, wheels would slip on soft soil, and the current may spike. Therefore, a limit on how much PWM can change per tick was set as a way to limit the sudden PWM changes. Two parameters were used to ensure that: P_{target} , the maximum duty cycle to reach, and T_{up} and T_{down} , the ramp up and ramp down time respectively in seconds. In manual mode, P_{target} is set to 30% with $T_{\text{up}} = 10$ s. There is no active T_{down} in manual mode as the main purpose of manual mode is accuracy and since the user is there all the time removing ramp down time did not seem as an issue.



Figure. 6.2a Manual Control Interface

In the case of guided mode, time constants are defined over the entire PWM range (0-100%) we have $T_{\text{up } 100} = 33.33$ s and $T_{\text{down } 100} = 23.33$ s. By setting them relative it's easier for the user to just change the PWM duty cycle (P_{target}) if needed and need not change the time constants. The PWM is capped at 30% as on uneven ground higher speeds caused the wheels to lose grip on turns, so 30% was taken as it had both coverage and control.

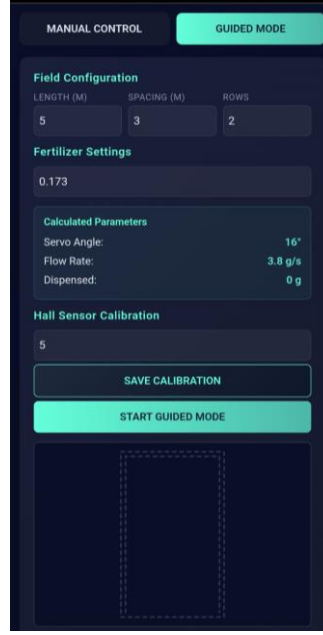


Fig. 6.2b Guided Control Interface

For proper direction reversal which was during turns, the approach function is used. These functions zeroing the current value before applying the new motion. This is to prevent instantaneous polarity change under load conditions and also not to trigger the safety condition present in motor controller which would not allow the motor to move.

6.2.2 Wait-for-Zero Synchronization

The ramp-down at 30% PWM takes about 7 seconds, so stopping the motors is not instantaneous. When the wheels were still decelerating during testing, the next sub-action would begin, and by the second or third row, the heading was obviously off. `Wait_for_zero()`, which halts execution and checks `currentl` and `currentr` every 20 ms until both fall below 1%, was the solution. In order to account for the worst-case scenario in which the mechanical load varies sufficiently on uneven ground that the deceleration is inconsistent, the timeout is set at 12 seconds instead of precisely 7.

6.2.3 Battery State Estimation

There isn't a separate coulomb-counter IC on the platform. So, the software sees it as a linear function of motor load:

$$I = 4.5 + \frac{|PWM_l + PWM_r|}{200 \times 26.8} \quad (6.1)$$

where 4.5 A is the baseline estimate no-load draw and 26.8 A represents the additional current drawn when both motors are at full load. To find the terminal voltage, take the resistive drop $I_{inst} \times R_{int}$ ($R_{int} = 0.08 \Omega$) from a three-segment open-circuit model:

25.6 V above 90% SoC, 24.0 V from 20% to 90%, and 23.8 V below 20%. It's not a very accurate measurement, but it's good enough for showing how long the battery will last and turning on a low-battery warning on the dashboard.

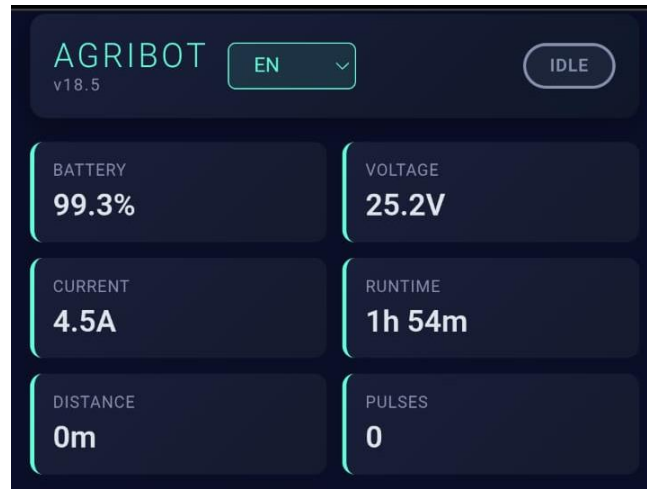


Fig. 6.2c Telemetry Display

6.3 Fertilizer Dispensing: Computation and Control

The fertilizer system uses a servo-actuated rotary valve, where the outlet hole size varies according to the angle of the shaft. The flow rate is described by a piecewise-linear function of the servo angle, with reference values set during initial commissioning. Nine operating points, covering the whole range from fully closed to fully open as shown in Table 6-1.

Table 6-1 Servo angle versus flow rate reference values

Angle (°)	Flow (g/s)
0	0.0
10	2.0
20	5.0
30	9.0
40	14.0
50	20.0
60	27.0
70	35.0
80	44.0

90	54.0
----	------

For angles between tabulated points, the flow rate is obtained by linear interpolation between the two nearest entries. For an angle θ lying between calibration points θ_1 and θ_2 :

$$Q(\theta) = Q_1 + (Q_2 - Q_1) \times \frac{\theta - \theta_1}{\theta_2 - \theta_1} \quad (6.2)$$

6.3.1 Automatic Angle Calculation

With the target fertilizer weight m (kg), row length L (m), and the number of rows R , the required servo angle is computed as follows. The robot moves in each row with an actual speed dependent on the guided mode duty cycle:

$$v_{actual} = v_{max} \times \frac{P_{guided}}{100} = 0.6 \times 0.30 = 0.18 \text{ m/s}$$

The fertilizer valve is activated only during the forward motion in the rows. Therefore, the total dispensing time is:

$$t_{total} = \frac{L \times R}{v_{actual}} \quad (6.3)$$

The required average flow rate is therefore:

$$Q_{required} = \frac{m \times 1000}{t_{total}} \quad (6.4)$$

The firmware cycles servo angles from 0° to 90° in 1° steps, assessing the interpolated flow rate at each step, and picks the smallest angle θ such that $Q(\theta) \geq Q_{required}$. A constant 2° margin is added to the obtained angle to guarantee the delivery of the target weight despite small mechanical differences in the valve.

6.4 Guided Operating Mode

The guided mode follows a semi-autonomous field coverage process for R rows of length L , with a spacing of S between the rows. This is designed in such a way that it has two operator confirmation points per row cycle, the one at the start of every row where the mulching sheet need to be anchored and the other where the mulching sheet has to be cut after completing a row and it needs to turn for the next row.

6.4.1 Mission Sequence

When a guided mode mission is initiated from the web interface, a dedicated mission thread is launched by firmware which is responsible for executing one complete row cycle at a time.

(a) Anchor confirmation

Execution begins in a paused state with wait_action set to "ANCHOR". The robot is at stationary until the operator manually positions and secures it at the beginning of the row. Once anchored, the operator confirms through the web interface. This action sets continue_flag under mutual exclusion, which the mission thread continuously monitors before advancing to the next stage.



The screenshot shows a web interface for a robot's mission sequence. At the top, there are two tabs: "MANUAL CONTROL" and "GUIDED MODE", with "GUIDED MODE" being the active tab. Below the tabs, the interface is divided into several sections: "Field Configuration" with input fields for "LENGTH (M)" (5), "SPACING (M)" (3), and "ROWS" (2); "Fertilizer Settings" with a value of 0.173; "Calculated Parameters" showing "Servo Angle: 16°", "Flow Rate: 3.8 g/s", and "Dispensed: 0 g"; and "Hall Sensor Calibration" with a value of 5. At the bottom, there are three buttons: "SAVE CALIBRATION", "ANCHORED - CONTINUE" (highlighted in green), and "ABORT MISSION" (highlighted in red). A dashed rectangular box is visible at the very bottom of the screen.

Fig. 6.4a Guided Mode Anchor Confirmation

(b) Row traversal

Row traversal. travel_distance(L) is called with dispense_fert = True. The fertilizer valve opens to the pre-calculated angle, both motors are set to the guided target, and the robot travels forward for the time $t = \frac{L}{V_{actual}}$. On completion, the motors are commanded to zero and wait_for_zero() blocks until full deceleration is confirmed. The fertilizer valve is then closed.

(c) Cut confirmation

The system transitions to another paused state with `wait_action = "CUT"`. The operator clears the mulching sheet at the end of the row and confirms completion via the interface. A `continue_flag` is generated when the complete button is pressed. Upon detection of `continue_flag`, the mission thread resumes to the next stage which is the turning stage.



Fig. 6.4b Guided Mode Cut Confirmation

(d) Repositioning turn

After completing a row and the `continue_flag` is given the robot performs a three-turn pattern to reposition the robot at the start of the next row (detailed in Section 6.4.2). After the turn sequence is over, `wait_action` is set back to "ANCHOR" so that the next row can start.

This cycle goes on for all the rows. When you finish the last row, the mission is over and the system goes back to the IDLE state.

6.4.2 Boustrophedon Coverage and Timed Pivot Turn

The robot moves across the field in a boustrophedon pattern, turning around at the end of each row instead of making a big U-turn. Moving between rows is handled in three phases: an in-place pivot, a lateral gap crossing, and a second pivot to restore heading.

In the first phase, one motor moves forward and the other moves backwards at 30% PWM for 5 seconds, which turns the motor about 90°. In reality, it took several test runs to get this pivot angle right because small changes in the ground's surface caused it to rotate slightly too much or too little, which had to be fixed by changing the PWM value instead of the duration. During the second phase, the robot moves forward across the space between the rows S. The time it takes to travel is given by $t = \frac{S}{v_{actual}}$ and the fertilizer valve held closed throughout. A second 5.0 s pivot in the same direction as first is done to restore the robot's forward alignment for the next row. At each phase transition, `wait_for_zero()` is called before issuing the next motor command, ensuring the robot has fully decelerated. The 0.4 s mechanical settling delay after the first pivot was something that had to be added after observing the chassis was still visibly oscillating when the lateral traverse started without which the second command would interrupt motion that had not actually stopped. The 0.6 s delay after the lateral traverse serves a similar purpose, though in testing it was less critical than the first.



Fig. 6.4c Guided Mode Boustrophedon Pattern

6.4.3 Hall-Sensor Distance Correction

Hall sensor is fixed to the shaft of the robot and is kept close to the rims of the wheel, where three strong neodymium magnets are placed one at each rim. When a magnet passes through the sensor the change in magnetic field is detected by the hall effect sensor thus registering a pulse. When the Hall-effect sensor is fitted, a correction pass follows each timed traversal. The expected pulse count for a target distance is:

$$N_{target} = L \times k_{hall} \quad (6.5)$$

where k_{hall} is the pulses-per-metre constant stored in the configuration file. If the accumulated count at the end of the timed run is less than N_{target} , the motors are re-engaged for a correction window of up to 3 seconds. This compensates for start-up slippage at low duty cycles, which can cause the robot to fall slightly short of the timed distance estimate. The time estimate was set at 3 seconds as to avoid a case where the sensor is faulty resulting in excessive forward motion due to incorrect pulse detection and have a fairly accurate system.

6.4.4 Watchdog and Emergency Stop

A communication watchdog monitors the interval since the last heartbeat packet received from the browser, which is transmitted every 1 second by the interface. If no packet arrives within `WATCHDOG_TO = 6.0` seconds while the robot is in guided motion with a non-zero motor target, the firmware infers a loss of operator connectivity. Motor targets are immediately cleared and the system transitions to SAFETY STOP, from which only a fresh mission start can recover. The 6-second threshold was a deliberate choice because tests showed that short Wi-Fi drops of 1–2 seconds were common. This meant that the system was hard to use because it had to stop for every small dropout. In guided mode, the watchdog is the main way to check if someone is still alive. In manual mode, each button press is proof that the operator is there.

The harder cutoff is handled by a different `emergency_stop_flag`. This flag completely avoids the ramp-down calculation and zeros both motor outputs within the same engine tick when the STOP button is pressed in the web interface. Since this is not the elegant deceleration used at the end of a traversal row, that distinction is important. It functions regardless of the ramp state or operating mode that the motors are in at the moment. As the name implies, this flag is only used in genuinely extremely difficult situations where the robot's path is incorrect, a motion fault is found, or the operation must be stopped right away for any reason.

6.5 Web Interface and Human-Machine Interaction

The Flask process runs the operator interface directly as a single-page web application server. Because all HTML, CSS, and JavaScript are integrated into the firmware source itself, there is no need for a separate build step, and the interface version is always identical to the firmware that is currently in use. There is no need to install any apps because operators connect via the robot's local Wi-Fi hotspot and view the interface in any standard browser. At render time, the Jinja2 template engine is used to replace all visible text strings that are

stored in a translation dictionary indexed by ISO 639-1 language codes. There are six languages available: Urdu, Telugu, Tamil, Hindi, Kannada, and English. Adding regional languages was intended to enable farmers to read all labels and warnings in their native tongue without help. As of right now, this feature is merely a proof of concept; additional languages could be added in later iterations without requiring structural modifications to the translation system. Both at login and via a dropdown on the main dashboard, users can select their preferred language, and the selection is maintained across page reloads through the Flask session.

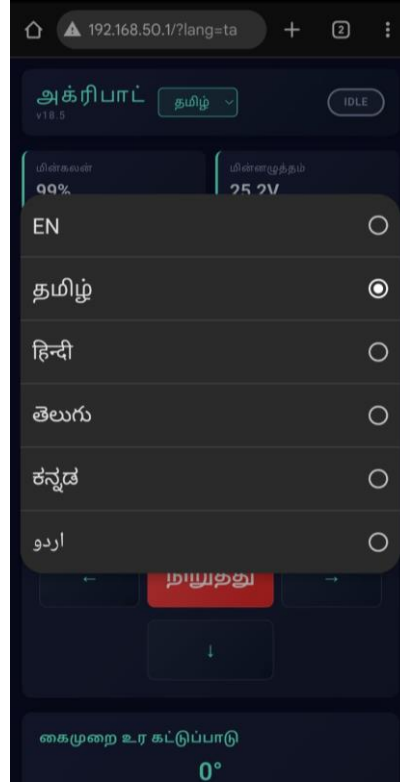


Fig. 6.5a Language Dropdown

Two tabbed panels comprise the dashboard. Every 200 ms, a telemetry bar at the top continuously polls `/api/data` to retrieve battery percentage, voltage, current, estimated runtime, distance, and Hall pulse count without requiring a page reload.

For directional input, the Manual Control tab employs a D-pad layout. Every 100 milliseconds, holding down any direction button retransmits the motor command; letting go of it instantly sends a STOP. The 100 ms retransmit loop was added clearly to make the button behave like a physical held-down control because during early testing, this repeat-on-hold behaviour was actually the most difficult to get right. A single tap would send one packet and the robot would continue to move, which was a risk.

The four mission inputs—L, S, R, and fertiliser mass—are entered into the Guided Mode tab. Without requiring any submissions, a live preview updates the calculated servo angle and flow rate as the operator types. This tab's mission control buttons modify based on the wait_action state, preventing the operator from selecting the next step before the robot has actually completed the current one. This prevented incorrect sequencing in previous versions prior to the introduction of this lock.

As the mission progresses, a canvas element on the same tab shades completed segments and creates a live schematic of the intended coverage path. After it's done, ReportLab creates a PDF session report that can be downloaded. Time stamps, total distance travelled, fertiliser dispensed, battery consumption, and a complete event log are all recorded in the report; during field testing, the event log proved especially helpful in identifying mid-mission challenges.

AgriBot Session Report	
Session Information	
Session ID:	4
Start Time:	2026-02-04 20:46:42
End Time:	2026-02-04 20:47:47
Status:	ABORTED
Field Configuration	
Length per Row:	3.0 meters
Row Spacing:	3.0 meters
Number of Rows:	2
Fertilizer Target:	0.173 kg
Performance Statistics	
Total Distance:	0.00 meters
Fertilizer Dispensed:	99.5 grams
Rows Completed:	1 / 2
Battery Start:	63.2%
Battery End:	62.2%
Battery Consumed:	1.0%
Event Log	
Time	Event
20:46:42	Session started
20:46:42	Waiting for initial anchor
20:46:51	Starting row 1/2
20:46:51	Fertilizer valve opened at 47°
20:46:56	Fertilizer valve closed
20:46:57	Completed row 1/2
20:46:57	Waiting for cutting
20:47:30	Executing turn
20:47:47	Waiting for anchoring
20:47:47	Session ended: ABORTED
Generated by AgriBot v18.5 on 2026-02-04 20:52:42	

Figure 6.5b PDF Session Report

7. Results

The robot was evaluated across several field runs covering four performance areas: fertilizer spread uniformity, bed profile consistency, path tracking accuracy, and operational time relative to manual methods.

Fertilizer wastage was reduced by 25–30% compared to manual spreading on equivalent plot lengths, where wastage rates ran at approximately 40%. The majority of that manual waste occurred at row starts, where operators tend to open the spreader before reaching a consistent walking pace, resulting in excess deposition in the first section of each row. The servo-gated dispensing mechanism avoids this by tying flow directly to the motion state. It was also observed during testing that hopper fill level influenced output at a fixed servo angle, so fill volume was standardized across all comparative runs to ensure measurement validity.

Bed height, width, and soil consistency were recorded across multiple passes and compared against a hand-formed reference bed. Robot-formed profiles showed approximately 20% greater consistency in cross-section. Manual bed formation performed adequately on initial rows but showed measurable drift in shoulder height by the third and fourth pass, attributed to cumulative operator fatigue. No such drift was observed in robotic passes. Soil structure across the formed beds appeared sufficient for root aeration and water retention, though controlled growth trials were outside the scope of this evaluation.

Field coverage reached 98% of the test plot area. Average lateral deviation from the planned path was ± 4 cm, with flatter sections consistently closer to ± 2 cm; a section with a slight gradient pushed deviation toward the upper end as the path calculation did not compensate for slope. For comparison, conventional tractor-based coverage of the same area requires approximately 35% greater travel distance, primarily due to wider turning radii at row ends, which increases both fuel consumption and total operation time.

8. Conclusion

The development of the Automated Combined Mulching Sheet Laying and Fertilizer Application Robot represents a big step towards the combining of two mechanisms in small-scale farming. This project has seen the complete development of the robot, where all the mechanical, electrical, software aspects of the robot were modeled, created, and integrated into one single system. This robot was found effective through various simulations and load conditions.

With the project complete, the results have verified the fact that the robotic system is compact, affordable, and effective at applying fertilizers and mulch in one go. Fabrication and hardware integration have been completed, and the robotic platform has performed well in different types of soil. The robotic system reduces the amount of manpower required and provides sustainable farming practices.

Thus, this project contributes to the field of agricultural automation by developing an automated solution that is scalable, adaptable, and affordable. The final product has the potential to increase the efficiency and optimize the use of resources for small and marginal farmers and make smart farming technology available to them.

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Publication

[1] Anjjai Rubhak T J, Adaikkappan S, Vishal Srivatsan S, and Peeyush K P, “Development of Automated Combined Mulching Sheet Laying and Fertilizer Application Robot” (Drafting)